

Navigation and Control Framework for Radar-Avoidant Quadrotor Flight in Pre-Mapped Environments

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- UAV missions in pre-mapped and radar-surveilled environments require:
 - safe navigation through cluttered areas,
 - avoidance of radar exposure regions,
 - accurate trajectory tracking under disturbances,
 - real-time feasibility with actuator constraints.
- Classical path planning alone does not guarantee trackable and robust motion.
- Pure control design alone does not explicitly handle global radar-avoidant routing.
- Hence, a **unified planning–control framework** is needed.

Main Idea and Contributions

- A hierarchical framework is proposed for **radar-avoidant quadrotor flight**.
- **RRT*** is employed to generate a collision-free and radar-aware global route.
- The waypoint sequence is converted into a **time-parameterized smooth reference trajectory**.
- The **outer loop** uses **IT-ASMC** to provide robust translational tracking.
- The **inner loop** uses **LPV-MPC** to regulate attitude under control constraints.
- A **CBF-based safety filter** is added to preserve safe forward invariance near danger regions.

Quadrotor Control Inputs

The four rotor angular speeds are denoted by

$$\Omega_i, \quad i = 1, 2, 3, 4,$$

where Ω_i is the angular speed of rotor i . Let C_T and C_Q denote the thrust and drag coefficients, respectively, and let l be the distance from the vehicle center to each rotor.

The aggregated control inputs are defined as

$$\begin{cases} U_1 = C_T (\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2), \\ U_2 = C_T (\Omega_2^2 - \Omega_4^2), \\ U_3 = C_T (\Omega_3^2 - \Omega_1^2), \\ U_4 = C_Q (\Omega_1^2 + \Omega_2^2 - \Omega_3^2 - \Omega_4^2). \end{cases} \quad (1)$$

- U_1 : total thrust,
- U_2 : roll moment,
- U_3 : pitch moment,
- U_4 : yaw moment.

Rotational Dynamics

Let (p, q, r) denote the body angular rates about the roll, pitch, and yaw axes, respectively. Let I_x , I_y , and I_z be the principal moments of inertia, J_r the total rotor inertia, and Ω the residual rotor-speed term.

Then, the rotational dynamics of the quadrotor are expressed as

$$\dot{p} = \frac{(I_y - I_z)qr - J_r q \Omega + U_2}{I_x}, \quad (2)$$

$$\dot{q} = \frac{(I_z - I_x)pr - J_r p \Omega + U_3}{I_y}, \quad (3)$$

$$\dot{r} = \frac{(I_x - I_y)pq + U_4}{I_z}. \quad (4)$$

These equations show that the attitude motion is nonlinear and coupled, which motivates the use of an inner-loop constrained controller based on LPV-MPC.

The translational motion in the inertial frame is given by

$$\ddot{X} = (\cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi) \frac{U_1}{m}, \quad (5)$$

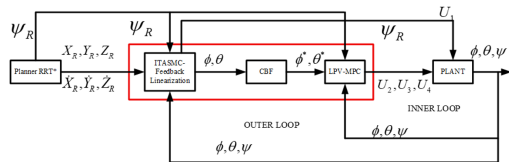
$$\ddot{Y} = (\cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi) \frac{U_1}{m}, \quad (6)$$

$$\ddot{Z} = -g + \cos \phi \cos \theta \frac{U_1}{m}. \quad (7)$$

- The translational subsystem is strongly coupled with attitude angles.
- This motivates a **cascade architecture**:
 - outer-loop position control,
 - inner-loop attitude control.

Overall Architecture

- The planner generates a radar-avoidant reference path.
- The path is smoothed into a time-parameterized trajectory.
- The **outer loop** computes virtual translational commands.
- The **CBF** modifies the nominal command only when safety is threatened.
- The **inner LPV-MPC** tracks attitude references while satisfying constraints.



Why LPV-MPC and IT-ASMC?

Inner loop: LPV-MPC

- The attitude subsystem is nonlinear and parameter-varying.
- LPV-MPC provides:
 - prediction-based attitude regulation,
 - explicit handling of input constraints,
 - improved feasibility in real-time implementation.

Outer loop: IT-ASMC

- The translational subsystem is affected by matched disturbances and uncertainties.
- IT-ASMC is introduced to achieve:
 - robust position tracking,
 - finite-time convergence behavior,
 - improved transient performance over long missions.

Inner-Loop LPV-MPC Formulation

The inner-loop attitude dynamics are written in LPV form:

$$\dot{x} = A(\sigma)x + B(\sigma)u, \quad y = Cx,$$

and discretized as

$$x_{k+1} = A(\sigma_k)x_k + B(\sigma_k)u_k.$$

The MPC solves a finite-horizon constrained optimization problem:

$$J = \frac{1}{2}e_Q^\top Q e_Q + \frac{1}{2} \sum_{k=0}^{N_p-1} \left(e_k^\top Q e_k + \Delta u_k^\top R \Delta u_k \right)$$

subject to

$$x_{k+1} = A(\sigma_k)x_k + B(\sigma_k)u_k, \quad \Delta u_{\min} \leq \Delta u_k \leq \Delta u_{\max}.$$

- This structure enables **high-frequency attitude tracking**.
- It also preserves **control feasibility** under actuator limits.

Outer-Loop IT-ASMC Design

The translational dynamics are rewritten as

$$\ddot{\xi} = v + d, \quad \xi = [X \ Y \ Z]^T,$$

where d denotes bounded matched disturbances.

Tracking errors are defined as

$$e_x = X - r_x, \quad e_y = Y - r_y, \quad e_z = Z - r_z.$$

The terminal integral sliding surface is selected as

$$s_i = \dot{e}_i + \alpha_i e_i + \beta_i \int_0^t e_i(\tau) d\tau + \gamma_i |e_i|^\mu \text{sign}(e_i), \quad i \in \{x, y, z\}.$$

The control law is decomposed as

$$v_i = v_i^{eq} + v_i^{sw},$$

where

- v_i^{eq} compensates nominal tracking dynamics,
- v_i^{sw} enhances robustness and finite-time error reduction.

CBF-Based Safety Filtering

To guarantee safety, a control barrier function is introduced over the safe set

$$\mathcal{C} = \{x \in \mathbb{R}^n \mid h(x) \geq 0\}.$$

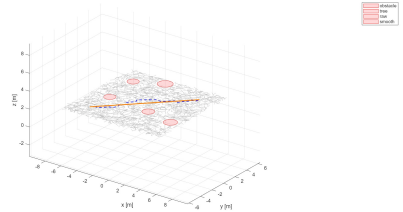
Safety is enforced by requiring

$$\dot{h}(x, u) + \alpha(h(x)) \geq 0.$$

- The CBF layer acts as a **real-time safety filter**.
- It only modifies the nominal control when the UAV approaches:
 - radar-threat regions,
 - obstacles,
 - unsafe forward states.
- Therefore, the controller remains as close as possible to the nominal tracking policy while preserving safety.

RRT*-Based Radar-Avoidant Planning

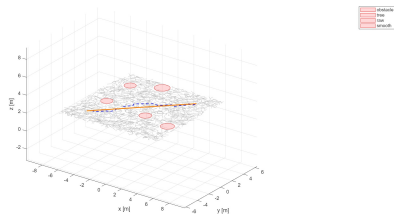
- **RRT*** is used in the planning layer to construct a feasible path from the initial to the target point.
- The planner accounts for:
 - obstacle avoidance,
 - radar-zone avoidance,
 - path optimality refinement.
- The resulting waypoint sequence is then smoothed into a trackable reference.



Simulation Setup and Evaluation Goal

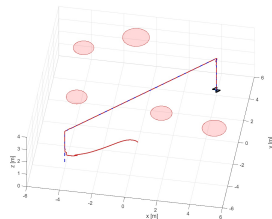
- The objective is to verify the proposed **planning–control integration**.
- The simulation assesses whether the UAV can:
 - generate a feasible radar-avoidant route,
 - follow the smoothed reference accurately,
 - maintain stable attitude regulation,
 - satisfy input constraints throughout the mission.
- The following figures report the planning result, 3D flight trajectory, axis-wise tracking, attitude response, and control signals.

Planned Route and 3D Flight Motion



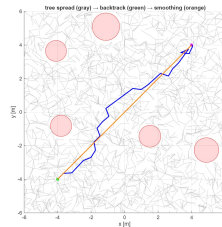
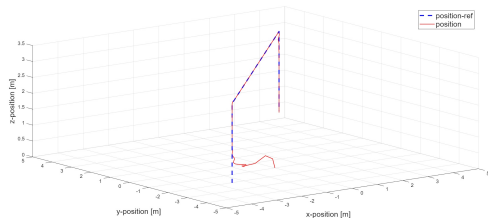
Planned radar-avoidant route generated by RRT*.

- The planner provides a collision-free and radar-aware route.
- The executed flight trajectory follows the intended corridor smoothly.



3D UAV motion along the generated trajectory.

Reference Tracking in the Horizontal Plane

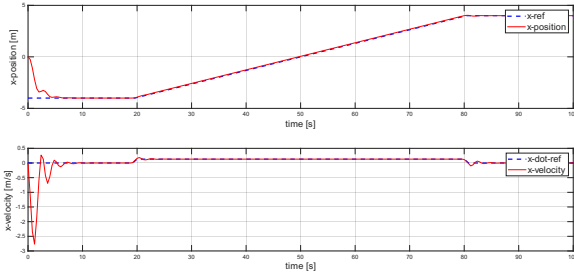


Actual and reference motion in the horizontal plane.

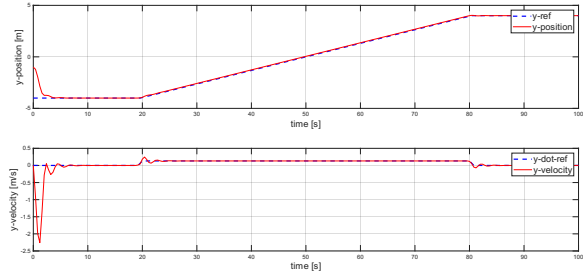
Spatial evolution of the trajectory in the environment.

- The actual path remains close to the planned/smoothed reference.
- The results confirm accurate path-following despite nonlinear coupled dynamics.

Position Tracking Along the x - and y -Axes



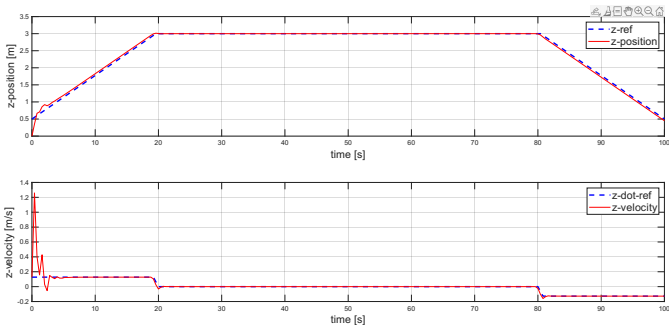
Tracking response along the x -axis.



Tracking response along the y -axis.

- Both horizontal position states converge toward their references with small steady-state error.
- The transient response remains well behaved during turning and path-transition phases.

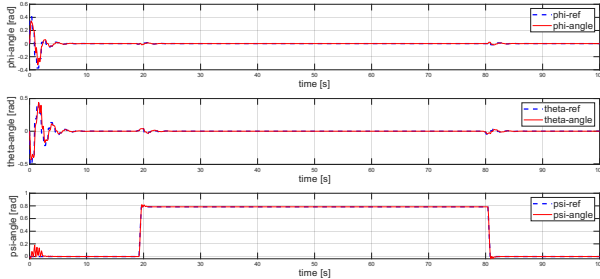
Position Tracking Along the z-Axis



Tracking response along the z-axis.

- The altitude tracks the desired profile smoothly.
- Vertical motion shows fast convergence and small oscillation.
- This indicates that the outer-loop controller produces consistent thrust-related commands for the inner-loop attitude/thrust regulation.

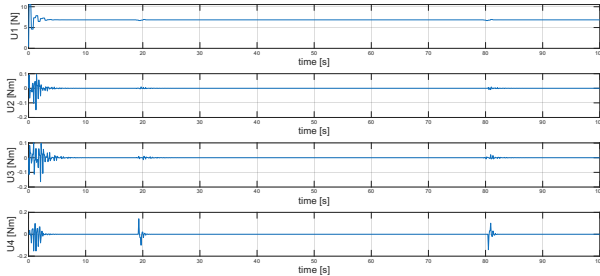
Attitude Tracking Performance



Roll, pitch, and yaw tracking performance.

- The attitude angles track their references rapidly.
- The responses are stable and sufficiently fast for trajectory execution.
- This confirms the effectiveness of the **LPV-MPC inner loop** for constrained attitude regulation.

Control Inputs



Control input evolution during the mission.

- The control signals remain bounded during the mission.
- No aggressive or unrealistic actuation is observed.
- This supports the feasibility of the proposed LPV-MPC-based inner-loop design.

The simulation results validate the proposed framework from both planning and control perspectives:

- **RRT*** successfully produces a radar-avoidant and obstacle-free path.
- The **trajectory smoothing stage** yields a trackable reference for the controller.
- The **IT-ASMC outer loop** ensures robust translational tracking.
- The **LPV-MPC inner loop** guarantees stable and feasible attitude regulation.
- The entire framework achieves accurate trajectory tracking with safe navigation in a constrained environment.

- A unified hierarchical navigation and control framework has been presented for radar-avoidant quadrotor flight in pre-mapped environments.
- **RRT*** handles global radar-aware path planning.
- **IT-ASMC** provides robust outer-loop translational tracking.
- **LPV-MPC** ensures constrained inner-loop attitude stabilization.
- **CBF** improves online safety by filtering unsafe commands.

The results demonstrate accurate tracking, control feasibility, and reliable radar-zone avoidance.

Thank you!